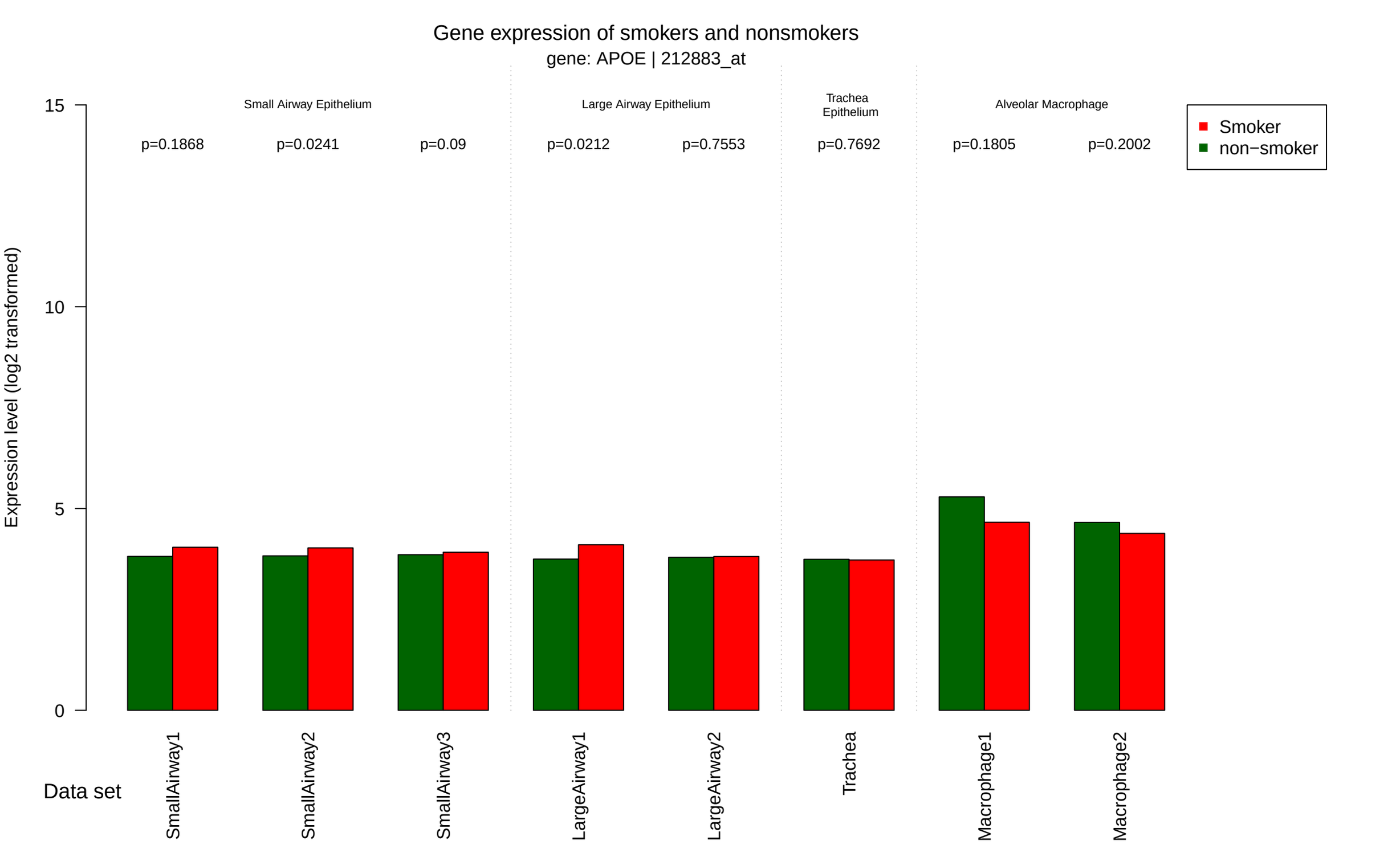
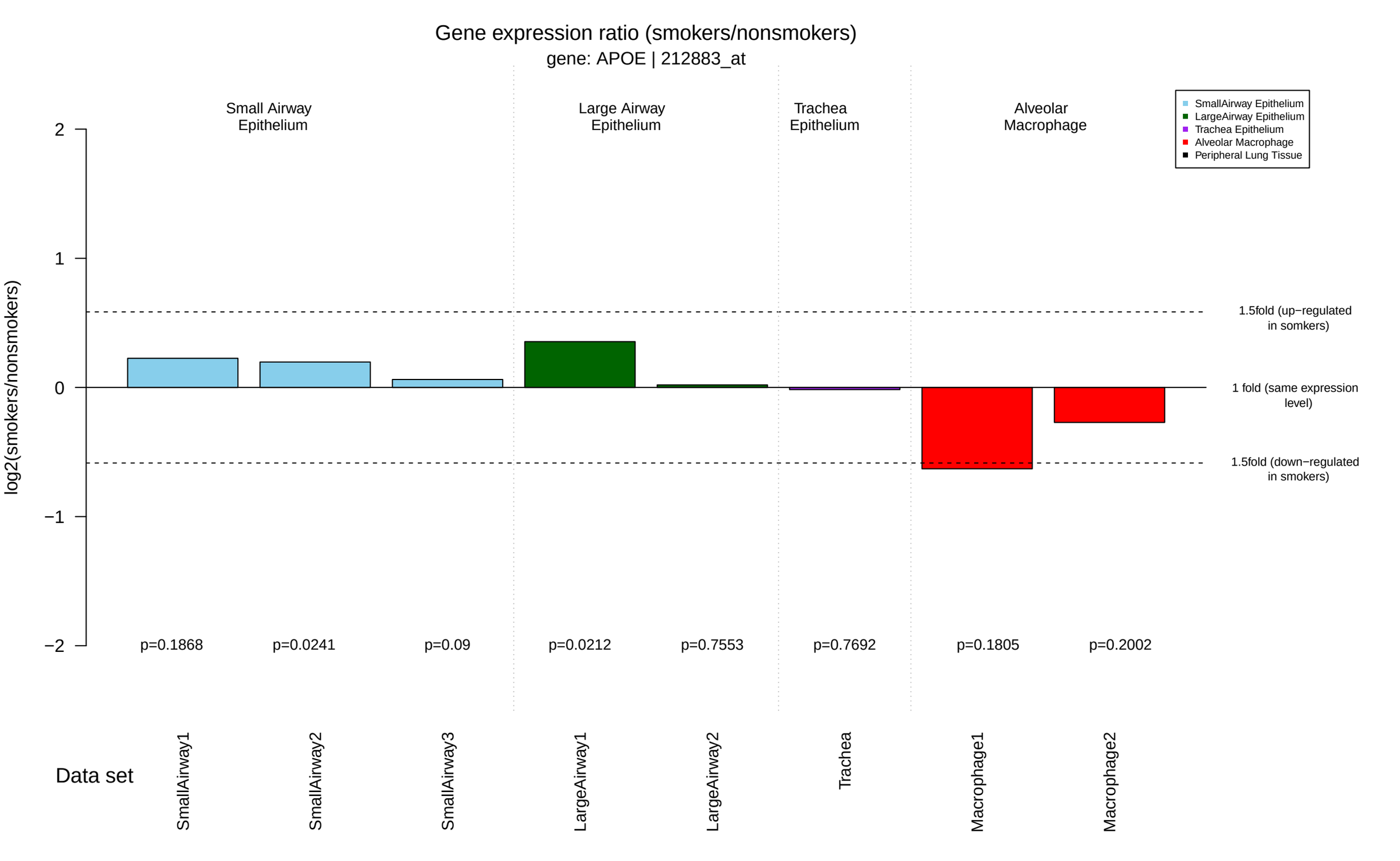
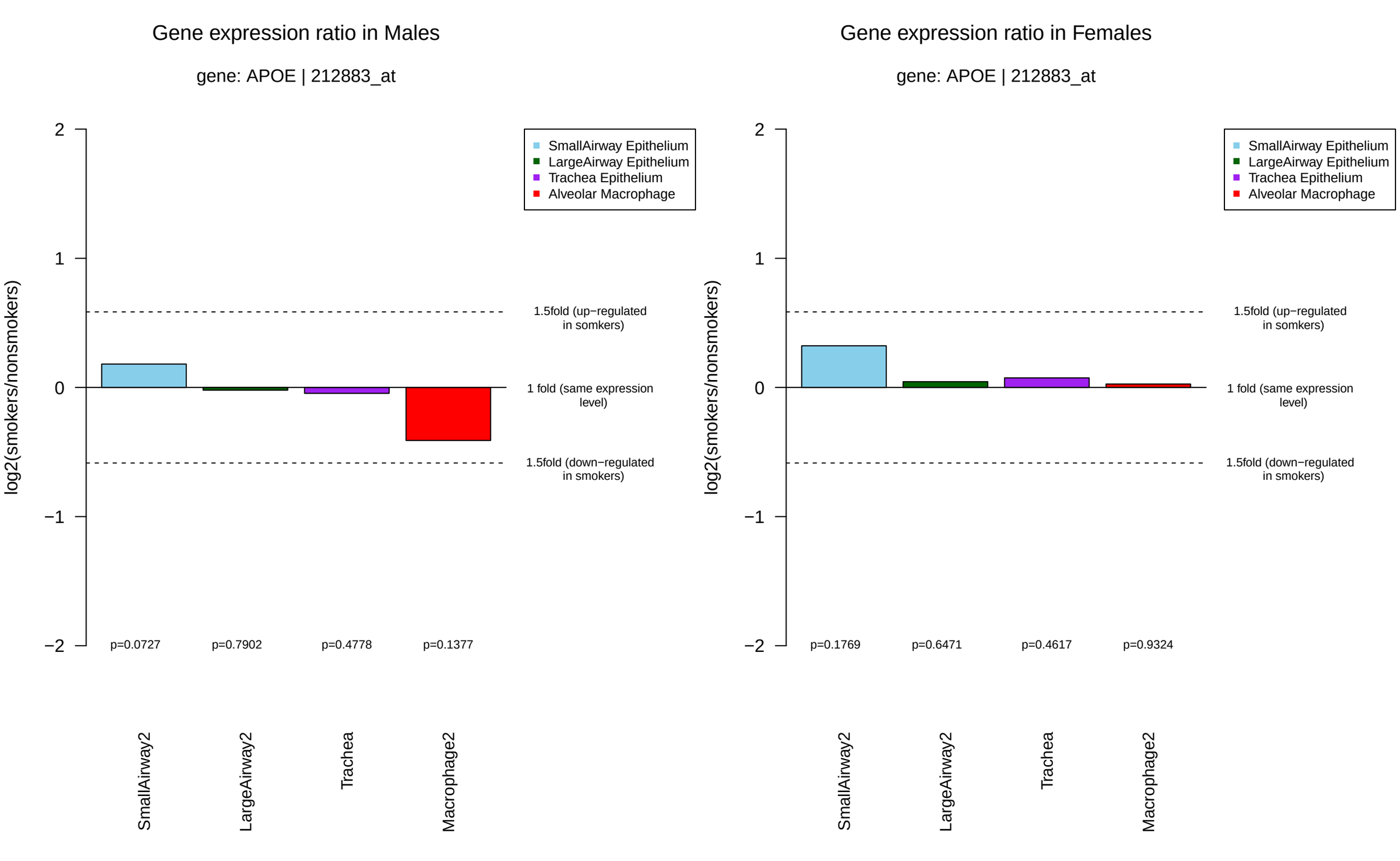
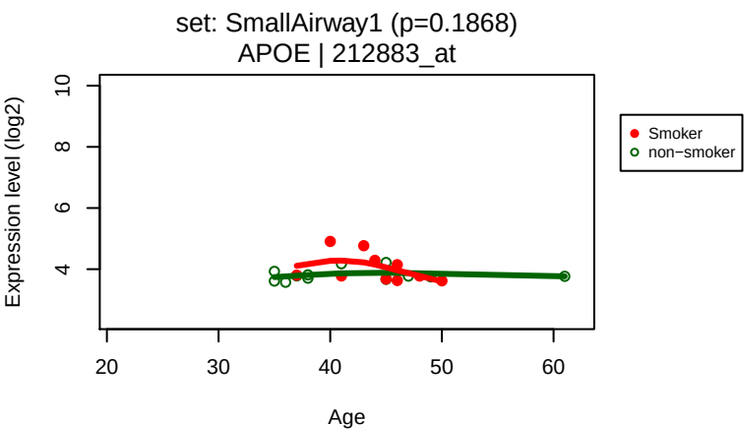
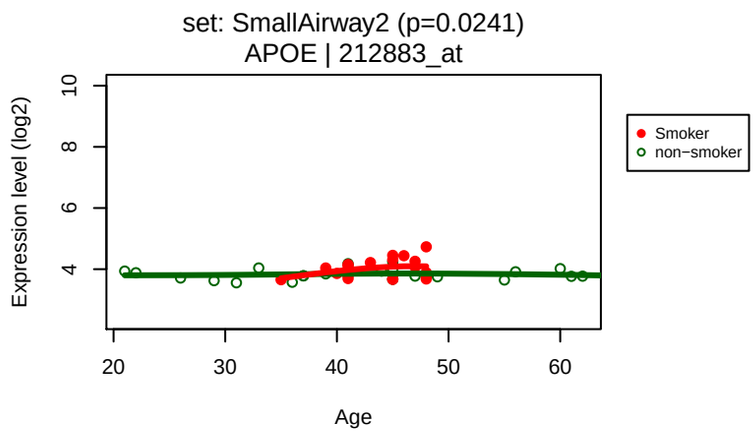
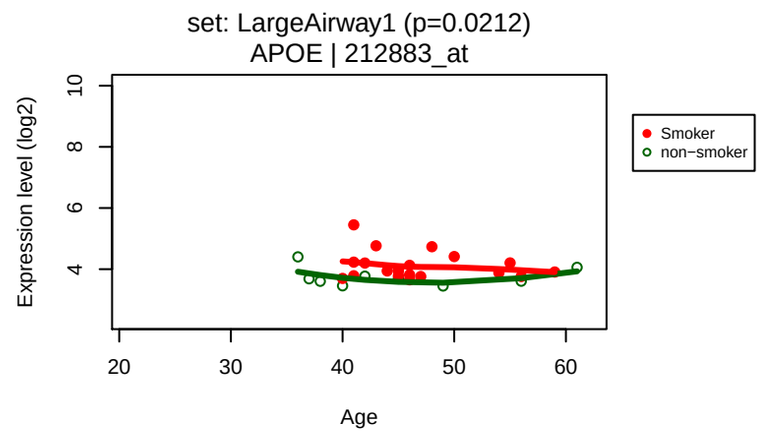

Gene expression of smokers and nonsmokers
gene: APOE | 212883_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1868
p=0.0241
p=0.09
p=0.0212
p=0.7553
p=0.7692
p=0.1805
p=0.2002
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

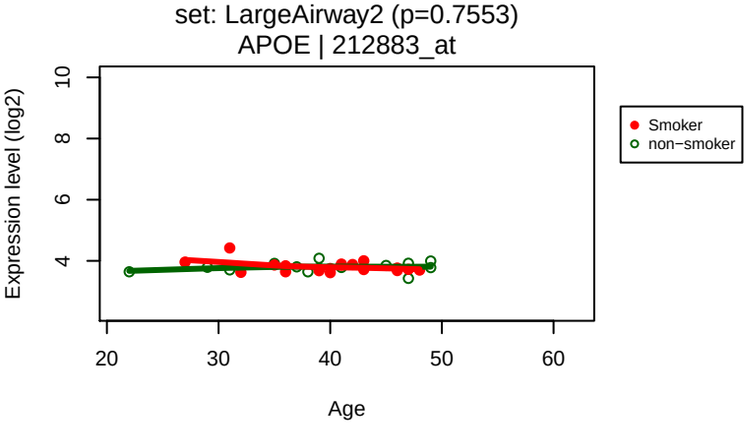

Gene expression ratio (smokers/nonsmokers)
gene: APOE | 212883_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1868
p=0.0241
p=0.09
p=0.0212
p=0.7553
p=0.7692
p=0.1805
p=0.2002
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

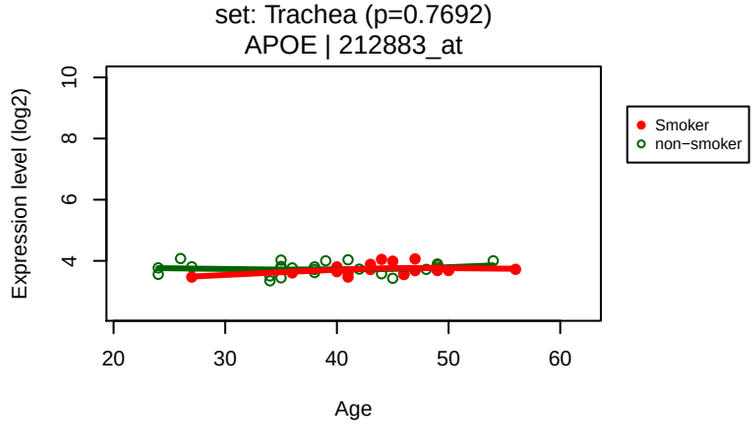

Gene expression ratio in Males
gene: APOE | 212883_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0727
p=0.7902
p=0.4778
p=0.1377
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: APOE | 212883_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1769
p=0.6471
p=0.4617
p=0.9324
SmallAirway2
LargeAirway2
Trachea
Macrophage2

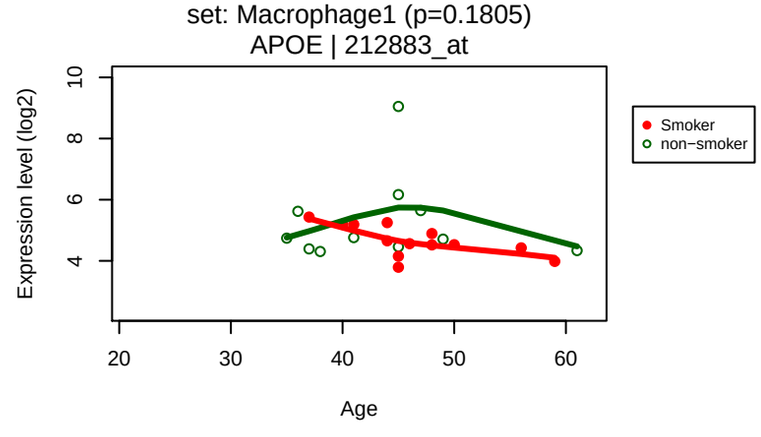

set: SmallAirway1 (p=0.1868)
APOE | 212883_at
Expression level (log2)
Age
Smoker
non−smoker

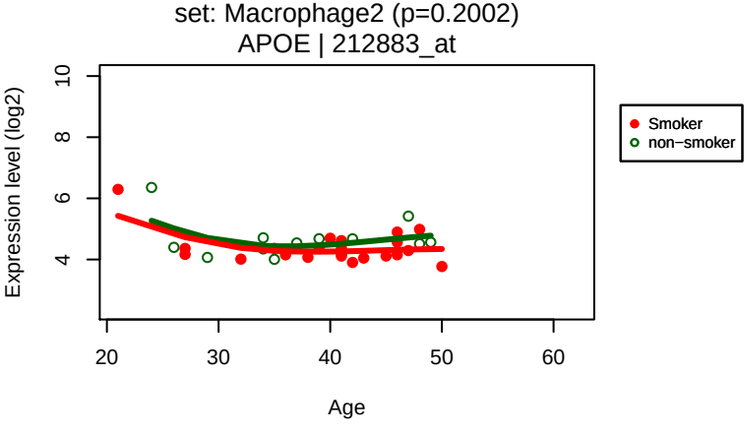

set: SmallAirway2 (p=0.0241)
APOE | 212883_at
Expression level (log2)
Age
Smoker
non−smoker

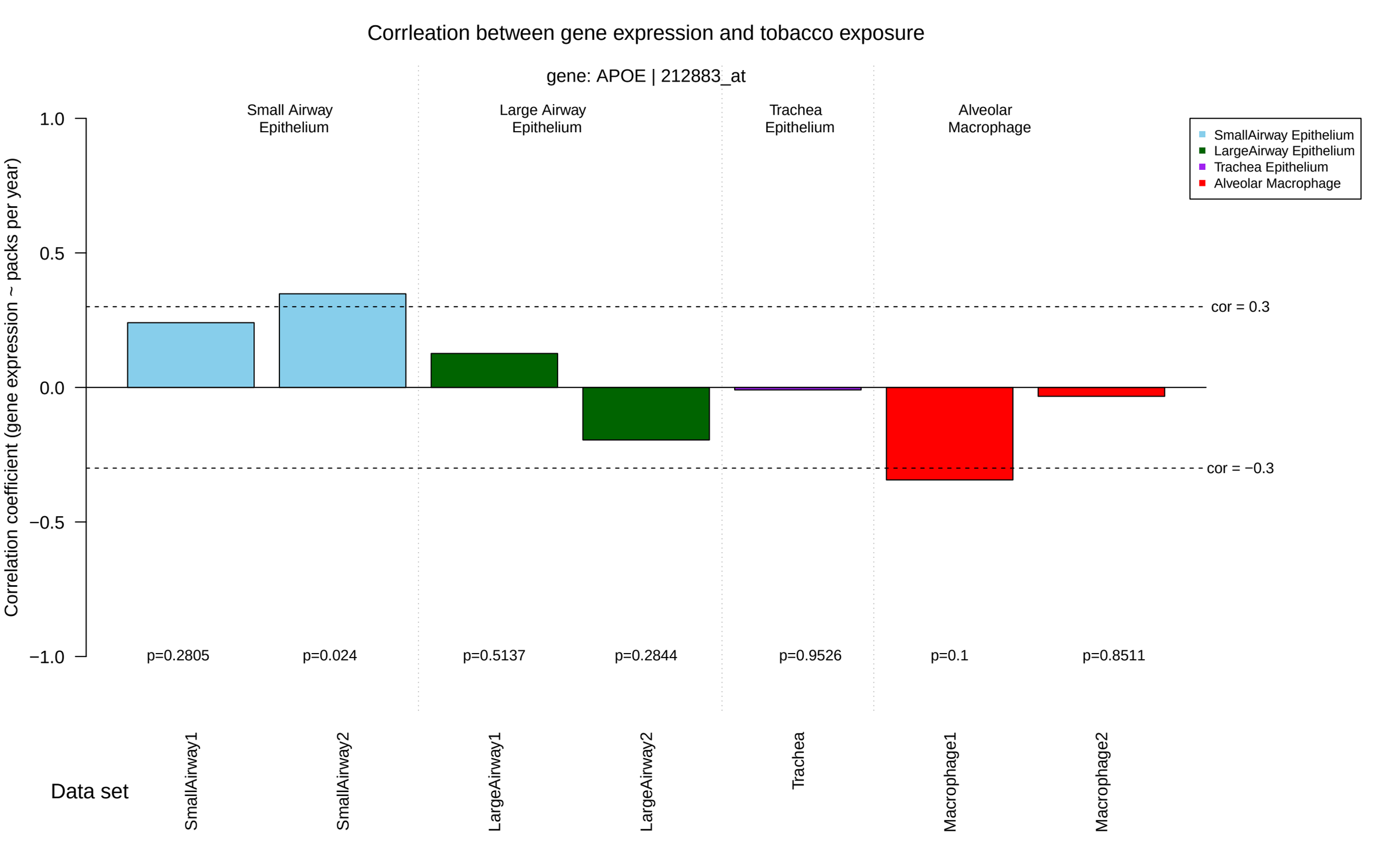

set: LargeAirway1 (p=0.0212)
APOE | 212883_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7553)
APOE | 212883_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7692)
APOE | 212883_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1805)
APOE | 212883_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2002)
APOE | 212883_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: APOE | 212883_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2805
p=0.024
p=0.5137
p=0.2844
p=0.9526
p=0.1
p=0.8511
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2